



Short-Term

Market Event Prediction

Using Stock Data

Move Tickers

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- **PIZZA**
- **META**
- **STOCK MARKET**

predicting stock market events (within the next 30 minutes)

For those who are ...

- When to invest more
- When to be cautious
- When withdraw their money

Move Tickers

- A modern software company
- Develops intuitive tools
- Expanding wealth management
- Leverages AI and machine learning to analyze market complexity

IMPORTANT STEPS

Data Engineering



- Engineered new features from existing data

EDA



- Explored data to check structure, correlations, and missing values.

Implement model



- Tested four classification models to find the one with the highest accuracy.

Launch



- Deployed the top model for real-time and batch predictions.

Source

- **Nasdaq**
- **BKNG**
- **Nasdaq 100**

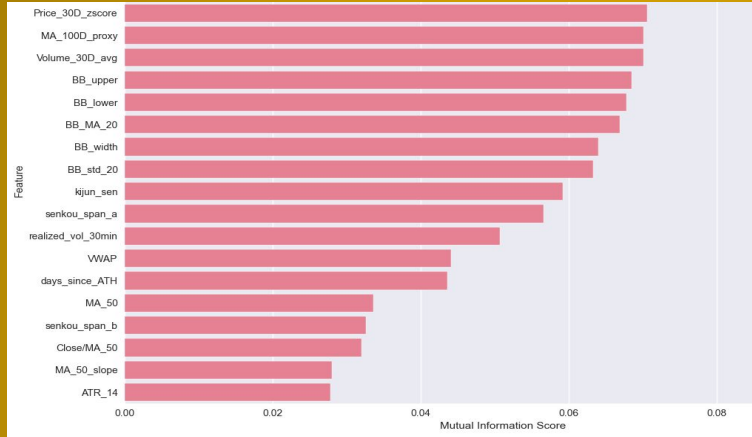


High-Frequency Feature Engineering for Anomaly Detection

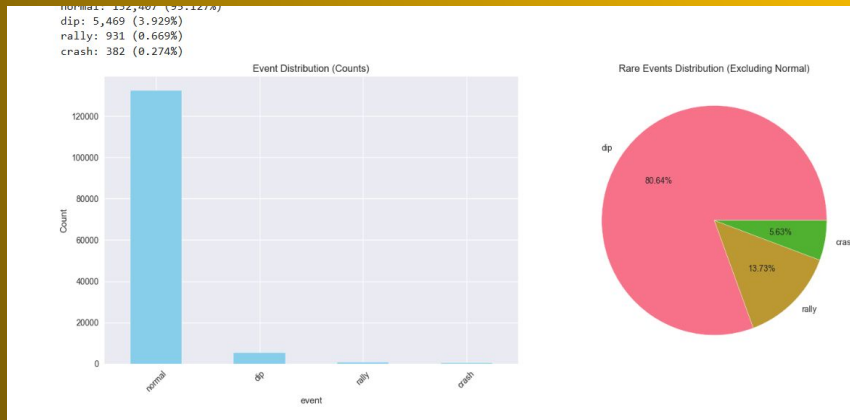
- Data Cleaning & Normalization
- Multi-Level Context Integration
- Feature Categories Created
 - a. Benchmarks: VWAP, Daily Moving Averages
 - b. Trend: 50 & 200-Minute Moving Averages
 - c. Momentum: RSI, MACD, Rate of Change (ROC)
 - d. Volatility: ATR, Realized Volatility
 - e. Temporal: Day of Week, Hour of Day



EDA



Top 10 Most Important Features:		
	Feature	MI_Score
42	tenkan_sen	0.097303
33	realized_vol_5min	0.080836
10	Price_30D_zscore	0.070669
9	MA_100D_proxy	0.070146
8	Volume_30D_avg	0.070138
37	BB_upper	0.068560
38	BB_lower	0.067793
35	BB_MA_20	0.066975
39	BB_width	0.064037
36	BB_std_20	0.063311



Number of missing business days: 215
 Number of Business Days in Dataset: 1785
 Percentage of missing days: 12.0 %



- **Applied time series analysis to financial data.**
- **Tested four models with two data balancing methods.**
- **Built a two-stage LSTM for anomaly detection and classification.**
- **Achieved reliable early warnings for market events.**

Ensuring Ethical and Regulatory Compliance in AI Systems

Ethical use and user awareness

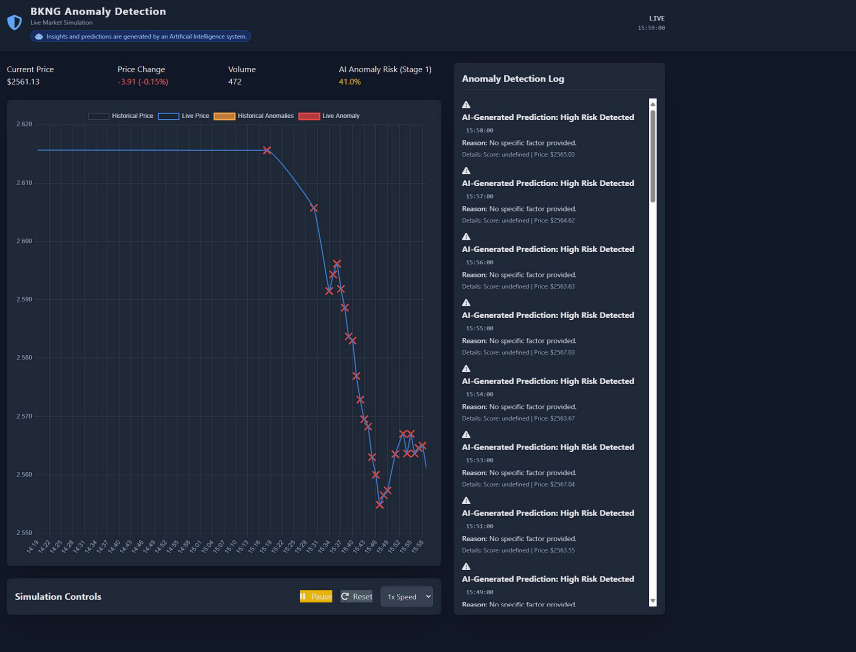
Training for AI literacy among users

Compliance with EU AI Act (Limited Risk)

Transparency in AI predictions

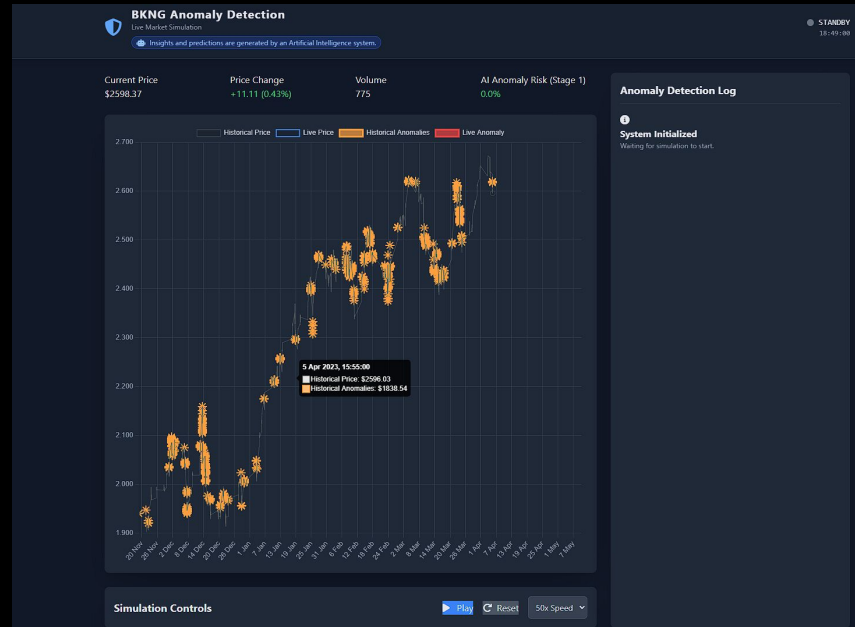
Clear user notifications for AI systems

GDPR data privacy and protection



- Shows real-time and historical price movements for BKNG stock.
- AI system detects and marks anomalies (potential risks) directly on the price chart.
- Displays a live log of AI-generated high-risk alerts, including timestamps.
- Includes key market information: current price, price change, volume, and AI anomaly risk score. (future additions)

UI/UX proof of concept



Questions?

**Thanks for
your attention**